

# Rızgar Ozan

Software engineer · games, tools and AI retrieval

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Software engineer in training who builds systems and the tools that prove they work. This summer our team shipped a local LLM stack, FastAPI/Docker data services and an Active Directory-backed internal portal at Barko Elektronik. On my own: a match-3 workbench with an engine-independent rules core, a Level Editor and a bot-driven difficulty curve (12,000 games in ~5 s), a Unity package (OpenUPM) that audits missing TextMeshPro glyphs across a whole project, and a reproducible Turkish retrieval benchmark. Comfortable in C# and Python; tests and measurements before claims.

## EDUCATION

**Hacettepe University, Ankara** — B.Sc. Computer Education and Instructional Technology (CEIT) Expected graduation June 2027 · GPA 3.00 / 4.00

## EXPERIENCE

**Software Engineering Intern** · Barko Elektronik, Ankara Jul 2026 – Aug 2026

- Two-person intern team. Deployed a fully local LLM stack on company hardware: benchmarked 7 open-weight models per use case and rolled out a company-wide assistant via Open WebUI with LDAP sign-in.
- Built FastAPI services, containerised with Docker, that pull data from Oracle and feed live Power BI dashboards, replacing manual reporting.
- Applied the local LLM to business workflows: CV analysis and categorisation, inconsistency detection in historical records, department-specific assistants.
- Unified department web apps into one internal portal on a shared design standard; integrated it with Active Directory over LDAP and implemented role-based access control.

## SELECTED PROJECTS

**Match3 Lab** — match-3 rules core, level format and bot simulator

C# (.NET, netstandard2.1) · Unity 6 · xUnit · GitHub Actions [github.com/RizgarOzan/match3-lab](https://github.com/RizgarOzan/match3-lab)

- Engine-independent, deterministic rules core (PCG32, event stream instead of callbacks) shared by a .NET CLI and a Unity presentation layer; 51 xUnit + 2 PlayMode tests; playable web demo.
- Parallel bot simulator measures level difficulty: 12,000 games in ~5 s on a laptop; difficulty curve, move-budget auto-tuner and CSV export.
- Found and fixed a bot 'foresight' bug (bot cloned the real RNG) that inflated win rates by up to 20 points; documented as an ADR.

**turkish-rag-eval** — Turkish retrieval evaluation harness

Python · BM25 · sentence-transformers · pytest · GitHub Actions [github.com/RizgarOzan/turkish-rag-eval](https://github.com/RizgarOzan/turkish-rag-eval)

- Reproducible benchmark: which parts of a RAG pipeline earn their cost in Turkish? 3 chunking strategies × 4 retrievers × 6 embedding models; 300-question gold set (58 human-written, 242 LLM-drafted and cross-labelled by a second model). Dataset public on Hugging Face, proposed as an MTEB task.
- Measured nDCG@10, Recall@5, MRR and CPU P95 latency. On 296 answerable questions: stemmed BM25 0.557, multilingual-e5-base 0.644, hybrid RRF 0.662; a 5-character prefix stemmer alone lifts BM25 by 23–29 % on the human-written set.
- Implemented abstention analysis (coverage vs. selective accuracy) and showed the top-1 margin on fused scores carries no signal; raw cosine does.

**tmp-glyph-audit** — Unity package that finds every TextMeshPro text your fonts cannot draw

C# · Unity Editor · UPM · OpenUPM · xUnit [github.com/RizgarOzan/tmp-glyph-audit](https://github.com/RizgarOzan/tmp-glyph-audit)

- Walks scenes, prefabs and runtime text through TMP's real fallback chain and reports missing glyphs per text/object/character; editor window plus CLI with CI exit codes.
- Pure C# core with 24 tests + 5 EditMode tests; published on OpenUPM (v0.2.0).

## OPEN SOURCE

**Open-source contributions** — 80 merged pull requests to 24 repositories I do not own (2026)

- ML / AI: embeddings-benchmark/mteb (18) and results (4), kornia/kornia (13: numerical edge cases in polynomial solvers and geometry, e.g. NaN on float16 or degenerate roots), HKUDS/LightRAG (4), docling (4), huggingface/sentence-transformers (1). Unity: CoplayDev/unity-mcp (7).
- Also mdn, radzen-blazor, atlassian-python-api, manim, pylint, openupm and others; 31 more are open and in review. Every PR reproduced first, with a regression test and full local CI.

## SKILLS

|                 |                                                                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages       | Python, C#, SQL, JavaScript/TypeScript (basic), HTML/CSS                                                                                                                                                |
| Engines & tools | FastAPI, Docker, Oracle / SQL, LDAP / Active Directory, Unity 6 (Editor tooling, UPM), .NET, xUnit, pytest, Git/GitHub Actions                                                                          |
| AI / data       | Local LLM deployment (Open WebUI), retrieval evaluation (nDCG, MRR), BM25 / dense / hybrid retrieval, RAG, sentence-transformers, Power BI                                                              |
| Spoken          | Turkish (native), English (B2, working proficiency)                                                                                                                                                     |
| Also shipped    | <b>Bilim Dedektifi</b> : released educational detective game, team course project at Hacettepe (Unity, C#). <a href="https://rizgarozan.itch.io/bilim-dedektifi">rizgarozan.itch.io/bilim-dedektifi</a> |